SUPPLY CHAIN CONTROL TOWER PROJECT

01_SQL_Analysis

Phase 2 — SQL Analysis Investigations (Pre-Dashboard SQL Investigations)

SQLite Database | Star Schema Data Model

```
SUPPLY CHAIN CONTROL TOWER PROJECT
01_SQL_Analysis.sql
Purpose:
  This file contains every SQL analysis (Phase 2: SQL Analysis) performed
  before building the Power BI dashboards, exactly as documented in the
  Supply Chain Control Tower Project Charter.
Phase Objective:
  Investigate root causes by connecting multiple datasets and answering
  business questions that cannot be easily solved through Excel alone.
  Develop data-driven supply chain insights by combining demand, inventory,
  supplier, transportation, warehouse, and forecasting data using SQL.
Database Engine : SQLite
Dataset         : 25,000 Orders | 8 Business Tables | 5,000 Shipments |
                   8 Warehouses | 498 SKUs
Tables Referenced in this file:
  Orders, Product, Inventory, Supplier, Supplier_Product_Map,
  Transportation, Warehouse, Forecast
Contents:
  Investigation 1  - Supplier Dependency Analysis
  Investigation 1A - Supplier Performance Risk Analysis
  Investigation 2  - Inventory Risk Analysis
  Investigation 3  - Forecast Accuracy Analysis
  Investigation 4  - Warehouse Inventory Risk Concentration Analysis
  Investigation 5  - Order Fulfillment Performance Analysis
  Investigation 6  - Transportation Carrier Performance Analysis
```

INVESTIGATION 1

Supplier Dependency Analysis

Business Question:
--   Which suppliers support the highest demand volume across the network?
Business Objective:
--   Identify supplier concentration risk and determine whether the business
--   is heavily dependent on a limited number of suppliers.
Tables Used:
--   Orders   (SKU_ID, Demand_Qty)   - customer demand volume
--   Supplier_Product_Map   (SKU_ID, Supplier_ID)   - maps products to suppliers
--   Supplier   (Supplier_ID, Supplier_Name)   - supplier master data
Data Relationship:
--   Orders -> SKU_ID -> Supplier_Product_Map -> Supplier_ID -> Supplier
Analytical Approach:
Measure : Demand_Qty
Dimension : Supplier_Name
Aggregation : SUM(Demand_Qty)
Reason : Determine total demand support by each supplier.
SQL Query:

```sql
SELECT
    s.Supplier_Name,
    SUM(o.Demand_Qty) AS Total_Demand
FROM Orders o
JOIN Supplier_Product_Map spm
    ON o.SKU_ID = spm.SKU_ID
JOIN Supplier s
    ON spm.Supplier_ID = s.Supplier_ID
GROUP BY s.Supplier_Name
ORDER BY Total_Demand DESC
LIMIT 3;
```

Expected Output:

```
--   Supplier_38 supports the highest demand volume (143,714 units),
--   followed by Supplier_49 (134,380 units) and Supplier_14 (129,265 units).
--   The top suppliers collectively support a significant portion of total
--   network demand.
```

Business Interpretation:

```
--   The business demonstrates supplier concentration risk, where a
--   significant portion of total demand is supported by a limited number
--   of suppliers. Any disruption affecting these suppliers could negatively
--   impact product availability, fill rate, and customer service
--   performance.
```

Recommendations: review supplier capacity and resilience for the

```
--   top-demand suppliers, monitor supplier concentration risk regularly,
--   evaluate alternate sourcing opportunities for critical SKUs, and
--   review inventory strategies for highly dependent supplier-product
--   combinations.
```

INVESTIGATION 1A

Supplier Performance Risk Analysis

```
Business Question:
--   Among the highest-demand suppliers, which suppliers present the
--   greatest operational risk based on delivery performance, lead time,
--   and quality metrics?
Business Objective:
--   Determine whether critical suppliers can reliably support customer
--   demand and identify potential risks to service levels.
Tables Used:
--   Orders                 (SKU_ID, Demand_Qty)
--   Supplier_Product_Map   (SKU_ID, Supplier_ID)
--   Supplier                (Supplier_ID, Supplier_Name, OTD_Pct,
--                             Lead_Time_Days, Defect_Rate)
Data Relationship:
--   Orders -> SKU_ID -> Supplier_Product_Map -> Supplier_ID -> Supplier
Analytical Approach:
Measure       : Demand_Qty
Aggregation   : SUM(Demand_Qty)
Supplier KPIs : OTD_Pct, Lead_Time_Days, Defect_Rate
Purpose       : Evaluate supplier reliability, responsiveness, and quality.
SQL Query:
```

```sql
SELECT
    s.Supplier_Name,
    SUM(o.Demand_Qty) AS Total_Demand,
    s.OTD_Pct,
    s.Lead_Time_Days,
    s.Defect_Rate
FROM Orders o
JOIN Supplier_Product_Map spm
    ON o.SKU_ID = spm.SKU_ID
JOIN Supplier s
    ON spm.Supplier_ID = s.Supplier_ID
GROUP BY
    s.Supplier_Name,
    s.OTD_Pct,
    s.Lead_Time_Days,
    s.Defect_Rate
ORDER BY Total_Demand DESC
LIMIT 3;
```

```
Expected Output:
--   Supplier_38 - Highest demand contributor; lowest OTD performance (65%);
--                 longest lead time (14 days); elevated defect rate (6.06%).
Assessment: highest operational risk among top-demand
--                 suppliers.
--   Supplier_49 - Second-highest demand contributor; highest OTD
--                 performance (75%); lowest defect rate (1.27%); moderate
--                 lead time (5 days). Assessment: strongest performing
--                 supplier among the top-demand contributors.
--   Supplier_14 - Third-highest demand contributor; shortest lead time
--                 (3 days); highest defect rate (7.31%). Assessment:
--                 quality-related supply chain risk.
Business Interpretation:
--   The organization is highly dependent on suppliers exhibiting relatively
--   weak delivery reliability and quality performance. Supplier_38
```

```
--   represents the most significant supply chain risk due to its
--   combination of high demand dependency, low OTD performance, long lead
--   time, and elevated defect rate. Any disruption could negatively impact
--   inventory availability, fill rate, and customer service levels.
Immediate actions: conduct a supplier performance review with
--   Supplier_38, investigate root causes of low OTD performance, review
--   lead-time reduction opportunities, and investigate quality issues
--   affecting Supplier_14. Medium-term actions: evaluate shifting demand
--   toward higher-performing suppliers where feasible, develop alternate
--   sourcing strategies for critical SKUs, review safety stock levels for
--   products supplied by Supplier_38, and implement supplier performance
--   monitoring dashboards.
```

INVESTIGATION 2

Inventory Risk Analysis

```
Business Question:
--   Which high-demand SKUs are operating below safety stock levels?
Business Objective:
--   Identify inventory shortage risks by combining customer demand and
--   inventory position data. The objective is to proactively identify
--   products that may cause stockouts, service-level failures, and
--   customer fulfillment issues.
Tables Used:
--   Orders     (SKU_ID, Demand_Qty)                       - customer demand
--                                                           volume per product
--   Inventory  (SKU_ID, Current_Stock, Safety_Stock)       - inventory
--                                                           availability and
--                                                           safety stock
--                                                           requirements
Data Relationship:
--   Orders -> SKU_ID -> Inventory
Analytical Challenge:
--   The Orders and Inventory tables operate at different levels of detail
--   (grain).
Orders Table Grain     : one row = one order transaction
Inventory Table Grain  : one row = one SKU in one warehouse
Desired Output Grain   : one row = one SKU
--   To avoid duplicate demand calculations, both datasets were aggregated
--   to SKU level before joining.
Analytical Approach:
--   Step 1 - Aggregate Demand    : SUM(Demand_Qty) per SKU
--   Step 2 - Aggregate Inventory : SUM(Current_Stock), SUM(Safety_Stock) per SKU
--   Step 3 - Calculate Inventory Gap:
--            Inventory_Gap = Total_Current_Stock - Total_Safety_Stock
--   Step 4 - Identify Inventory Risk:
--            Filter WHERE Total_Current_Stock < Total_Safety_Stock
SQL Query:
```

```sql
SELECT
    o.SKU_ID,
    o.Total_Demand,
    i.Total_Current_Stock,
    i.Total_Safety_Stock,
    (i.Total_Current_Stock - i.Total_Safety_Stock) AS Inventory_Gap
FROM
    (
        SELECT
            SKU_ID,
            SUM(Demand_Qty) AS Total_Demand
        FROM Orders
        GROUP BY SKU_ID
    ) o
LEFT JOIN
    (
        SELECT
            SKU_ID,
            SUM(Current_Stock) AS Total_Current_Stock,
            SUM(Safety_Stock) AS Total_Safety_Stock
        FROM Inventory
```

```
        GROUP BY SKU_ID
    ) i
ON o.SKU_ID = i.SKU_ID
WHERE i.Total_Current_Stock < i.Total_Safety_Stock
ORDER BY o.Total_Demand DESC;
```

```
Expected Output:
--   SKU0476 - Highest demand among inventory-risk SKUs; current stock of
--             185 units versus safety stock requirement of 336 units;
--             inventory deficit of 151 units. Assessment: highest
--             inventory exposure within the network.
--   SKU0107 - Demand volume of 8,906 units; current inventory reduced to
--             only 1 unit; inventory deficit of 50 units. Assessment:
--             immediate stockout risk due to critically low inventory
--             availability.
Network-Level Observation: the Inventory table contains 498 unique
--   SKUs; only 2 SKUs were identified below safety stock levels.
--   Inventory Health Rate = (496 / 498) x 100 = 99.6%.
Note: This investigation aggregates inventory at SKU level across all warehouses. Therefore only
2 network-level SKUs remain below total safety stock, whereas the Power BI dashboards analyze
warehouse-level inventory and report 234 warehouse-SKU risk records.
Business Interpretation:
--   The overall inventory network appears healthy, with approximately
--   99.6% of SKUs operating at or above safety stock levels. However,
--   SKU0476 represents a significant inventory exposure due to high demand
--   and a substantial inventory deficit, and SKU0107 faces immediate
--   stockout risk because inventory availability is nearly depleted.
--   Failure to replenish these products could negatively impact fill
--   rate, service level, customer satisfaction, and revenue performance.
Immediate actions: prioritize replenishment for SKU0107, review
--   inventory planning parameters for SKU0476, and expedite replenishment
--   orders for both SKUs. Medium-term actions: review safety stock
--   policies for critical products, implement automated inventory-risk
--   alerts, improve inventory monitoring through Control Tower dashboards,
--   and conduct periodic inventory health reviews for high-demand SKUs.
```

INVESTIGATION 3

Forecast Accuracy Analysis

```
Business Question:
--   Which SKUs have the highest forecast error?
Business Objective:
--   Identify products with poor forecast accuracy to improve demand
--   planning, reduce stockouts, minimize excess inventory, and optimize
--   inventory investment.
Tables Used:
--   Forecast  (SKU_ID, Forecast_Qty, Actual_Demand)
--              - Forecast_Qty represents planned customer demand.
--              - Actual_Demand represents realized customer demand.
--              - Enables measurement of planning accuracy.
Data Relationship:
Only a single table is required: Forecast.
Analytical Approach:
Desired Output Grain : one row = one SKU
--   The Forecast table contains multiple records by SKU, warehouse, and
--   month; therefore the data is aggregated to SKU level to evaluate
--   overall planning accuracy for each product.
--   Measure 1 - Forecast_Qty      : SUM(Forecast_Qty)
--                                   Calculate the total planned demand per SKU.
--   Measure 2 - Actual_Demand     : SUM(Actual_Demand)
--                                   Calculate the total actual customer demand
--                                   per SKU.
--   Measure 3 - Forecast Error    : Total_Actual_Demand - Total_Forecast
--                                   Positive = Underforecasting (Actual > Forecast)
--                                   Negative = Overforecasting (Forecast > Actual)
--   Measure 4 - Absolute Forecast Error : ABS(Forecast Error)
--                                   Measures the magnitude of forecasting
--                                   error regardless of direction and enables
--                                   ranking of the largest planning deviations.
SQL Query:
```

```sql
SELECT
    SKU_ID,
    SUM(Forecast_Qty) AS Total_Forecast,
    SUM(Actual_Demand) AS Total_Actual_Demand,
    SUM(Actual_Demand) - SUM(Forecast_Qty) AS Forecast_Error,
    ABS(SUM(Actual_Demand) - SUM(Forecast_Qty)) AS Absolute_Forecast_Error
FROM Forecast
GROUP BY SKU_ID
ORDER BY Absolute_Forecast_Error DESC
LIMIT 5;
```

```
Expected Output:
--   SKU0308 - Largest forecast deviation in the network; demand exceeded
--             forecast by 1,881 units. Assessment: highest underforecast
--             risk, increasing the likelihood of stockouts and
--             service-level failures.
--   SKU0235 - Demand exceeded forecast by 1,737 units. Assessment:
--             additional underforecasting issue requiring review of
--             planning assumptions.
--   SKU0186 - Forecast exceeded actual demand by 1,790 units. Assessment:
--             largest overforecasting case, indicating excess inventory
--             exposure.
```

```
--   SKU0472 - Forecast exceeded actual demand by 1,644 units. Assessment:
--             potential inventory carrying cost and warehouse utilization
--             concerns.
--   SKU0464 - Forecast exceeded actual demand by 1,620 units. Assessment:
--             significant planning deviation results in excess inventory
--             risk.
```

Business Interpretation:

```
--   The analysis identified both underforecasting and overforecasting
--   across critical SKUs. Underforecasting may lead to stockouts, reduced
--   service level, lost sales, and emergency replenishment.
--   Overforecasting may lead to excess inventory, higher inventory
--   carrying costs, poor warehouse space utilization, and working capital
--   tied up in inventory. The presence of significant forecast deviations
--   in both directions indicates opportunities to improve forecasting
--   accuracy and demand planning processes.
```

Immediate actions: review planning assumptions for the top

```
--   forecast-error SKUs, investigate root causes behind significant
--   forecasting deviations, and adjust future forecasts using historical
--   forecast performance. Medium-term actions: establish forecast accuracy
--   as a recurring planning KPI, improve collaboration between Sales,
--   Demand Planning, and Supply Planning, incorporate historical forecast
--   error into future planning cycles, and develop Power BI dashboards to
--   monitor forecast accuracy and forecast bias over time.
```

INVESTIGATION 4

Warehouse Inventory Risk Concentration Analysis

```
Business Question:
--   Which warehouses contain the highest concentration of inventory-risk
--   SKUs while supporting high customer demand?
Business Objective:
--   Identify warehouses that contain the greatest number of inventory-risk
--   SKUs while supporting significant customer demand. This enables the
--   supply chain team to prioritize warehouse-level replenishment efforts
--   and reduce the risk of stockouts affecting customer service.
Tables Used:
--   Orders     (Warehouse_ID, Demand_Qty)
--                - Provides customer demand handled by each warehouse.
--   Inventory  (Warehouse_ID, SKU_ID, Current_Stock, Safety_Stock)
--                - Identifies inventory-risk SKUs by comparing current
--                  inventory against safety stock.
Data Relationship:
--   Orders -> Warehouse_ID -> Inventory
Table Grain:
Orders    : one row = one order transaction
Inventory : one row = one SKU in one warehouse
Desired Output Grain : one row = one warehouse
Analytical Approach:
--   Measure 1 - Demand_Qty : SUM(Demand_Qty)
--               Calculate the total customer demand handled by each warehouse.
--   Measure 2 - Risk SKU   : Business Rule Current_Stock < Safety_Stock
--               Aggregation COUNT(SKU_ID)
--               Count the number of inventory-risk SKUs within each warehouse.
SQL Query:
```

```sql
SELECT
    o.Warehouse_ID,
    o.Total_Demand,
    COALESCE(i.Risk_SKU_Count, 0) AS Risk_SKU_Count
FROM
    (
        SELECT
            Warehouse_ID,
            SUM(Demand_Qty) AS Total_Demand
        FROM Orders
        GROUP BY Warehouse_ID
    ) o
LEFT JOIN
    (
        SELECT
            Warehouse_ID,
            COUNT(SKU_ID) AS Risk_SKU_Count
        FROM Inventory
        WHERE Current_Stock < Safety_Stock
        GROUP BY Warehouse_ID
    ) i
ON o.Warehouse_ID = i.Warehouse_ID
ORDER BY
    o.Total_Demand DESC;
```

```
Expected Output:
--   Warehouses ranked by total customer demand alongside the number of
```

```
--   inventory-risk SKUs. WH7 had the highest Risk SKU Count (53), while
--   WH5 processed the highest demand. The analysis enables planners to
--   identify operational hotspots where inventory shortages are
--   concentrated despite high warehouse activity.
```

Business Interpretation:

```
--   Traditional warehouse-level inventory aggregation indicated that all
--   warehouses held inventory above their overall safety stock levels.
--   However, this masked SKU-level shortages within individual
--   warehouses. By analyzing the concentration of inventory-risk SKUs
--   instead of total warehouse inventory, the Control Tower provides a
--   more accurate representation of operational inventory risk. This
--   allows planners to prioritize replenishment efforts toward warehouses
--   where multiple critical SKUs are below safety stock, improving
--   inventory visibility and reducing potential stockouts.
```

Recommendations: prioritize replenishment planning for warehouses

```
--   with the highest number of inventory-risk SKUs, review inventory
--   allocation policies to reduce SKU-level shortages within warehouses,
--   investigate recurring inventory-risk patterns to identify root causes
--   such as inaccurate forecasting, replenishment delays, or supplier
--   issues, and develop warehouse-level dashboards to continuously
--   monitor inventory-risk concentration.
```

INVESTIGATION 5

Order Fulfillment Performance Analysis

```
Business Question:
--   What percentage of customer orders are fulfilled with the complete
--   quantity demanded?
Business Objective:
--   Evaluate customer order fulfillment performance by measuring the
--   percentage of orders where the delivered quantity meets or exceeds
--   the quantity demanded. The analysis helps identify service-level gaps
--   caused by incomplete order fulfillment and provides a foundation for
--   investigating potential issues related to inventory availability,
--   forecasting, warehouse operations, and supplier performance.
Tables Used:
--   Orders  (Order_ID, Demand_Qty, Delivered_Qty)
--            - Order_ID identifies individual customer orders.
--            - Demand_Qty represents the quantity requested by the customer.
--            - Delivered_Qty represents the quantity actually delivered.
Table Grain:
--   One row represents one customer order transaction.
Desired Output Grain:
--   One row representing overall network-level In-Full order fulfillment
--   performance.
Analytical Approach:
--   Measure 1 - Total Orders    : COUNT(Order_ID)
--               Calculate the total number of customer orders evaluated.
--   Measure 2 - In-Full Orders  : Business Rule Delivered_Qty >= Demand_Qty
--               An order is classified as In Full when the delivered
--               quantity is equal to or greater than the quantity demanded.
Aggregation: SUM(CASE WHEN Delivered_Qty >= Demand_Qty
--               THEN 1 ELSE 0 END). The CASE WHEN statement evaluates each
--               order individually (1 = In Full, 0 = Not In Full); SUM()
--               then adds all orders classified as 1 to calculate the
--               total number of In-Full orders.
--   Measure 3 - In-Full Rate    : (In-Full Orders / Total Orders) x 100
--               Measures the percentage of customer orders where the
--               complete requested quantity was delivered.
SQL Query:
```

```sql
SELECT
    COUNT(Order_ID) AS Total_Orders,
    SUM(
        CASE
            WHEN Delivered_Qty >= Demand_Qty THEN 1
            ELSE 0
        END
    ) AS In_Full_Orders,
    ROUND(
        100.0 *
        SUM(
            CASE
                WHEN Delivered_Qty >= Demand_Qty THEN 1
                ELSE 0
            END
        )
        / COUNT(Order_ID),
        2
```

```sql
    ) AS In_Full_Rate_Pct
FROM Orders;
```

Expected Output:
```
--   Out of 25,000 customer orders, 12,587 orders were fulfilled in full,
--   resulting in an overall In-Full Rate of 50.35%.
```
Additional calculation: Not-In-Full Orders = 12,413;
```
--   Not-In-Full Rate = 49.65%.
```
OTIF Data Limitation:
```
--   The original objective was to evaluate OTIF (On Time In Full)
--   performance. However, true OTIF cannot be calculated using the
--   currently available dataset. The dataset provides Demand_Qty and
--   Delivered_Qty, which allows calculation of the In-Full component.
--   However, the dataset does not contain sufficient order-level
--   information such as Promised/Required Delivery Date or Actual
--   Delivery Date. Therefore, the On-Time component cannot be reliably
--   determined. The Transportation table contains Delivery_Days, but
--   without a defined committed delivery time and a direct
--   Order-to-Shipment relationship, it cannot be used to calculate true
--   order-level OTIF. Therefore, the analysis focuses on In-Full
--   performance rather than presenting an inaccurate OTIF KPI.
```
Business Interpretation:
```
--   An In-Full Rate of 50.35% indicates that nearly half of customer
--   orders were not fulfilled with the complete quantity requested. Poor
--   In-Full performance can potentially contribute to reduced customer
--   service levels, increased backorders or partial deliveries, lower
--   customer satisfaction, additional transportation or handling
--   requirements for remaining quantities, and potential loss of future
--   customer demand. The result highlights the need for deeper analysis
--   to identify where Not-In-Full orders are concentrated and determine
--   the underlying operational causes.
```
Immediate actions: identify the SKUs contributing most frequently to
```
--   Not-In-Full orders, investigate whether Not-In-Full orders are
--   concentrated within specific warehouses, review inventory
--   availability for frequently under-fulfilled SKUs, and compare
--   under-fulfilled SKUs against inventory-risk SKUs identified in
--   Investigation 2. Further analysis (incorporated into the Power BI
--   dashboards where applicable): analyze In-Full performance by
--   warehouse, by customer region, and by SKU/product category; and
--   investigate relationships between forecast errors and under-fulfilled
--   orders, and whether warehouses with higher Risk SKU Counts also
--   experience lower In-Full performance.
```

INVESTIGATION 6

Transportation Carrier Performance Analysis

```
Business Question:
--   How do transportation carriers perform in terms of delivery speed and
--   freight cost, and are there route-specific differences that should
--   influence carrier selection?
Business Objective:
--   Evaluate transportation carrier performance by comparing shipment
--   volume, average delivery time, and freight cost. The objective is to
--   identify potential cost-service trade-offs between carriers and
--   determine whether carrier performance varies across different
--   transportation lanes. This analysis supports more informed carrier
--   allocation decisions based on shipment urgency, transportation cost,
--   and route-specific performance.
Tables Used:
--   Transportation (Shipment_ID, Carrier, Origin, Destination,
--                   Freight_Cost, Delivery_Days)
--                   - Shipment_ID identifies individual shipments.
--                   - Carrier identifies the transportation service provider.
--                   - Origin identifies the shipment starting location.
--                   - Destination identifies the shipment delivery location.
--                   - Freight_Cost represents transportation cost.
--                   - Delivery_Days represents the number of days required
--                     for delivery.
Table Grain:
--   One row represents one shipment transaction.
```

Analysis 1: Overall Carrier Performance

```
Business Question:
--   How do carriers compare across the overall transportation network in
--   terms of shipment volume, average delivery time, and freight cost?
Desired Output Grain:
--   One row represents one carrier.
Analytical Approach:
--   Measure 1 - Total Shipments      : COUNT(Shipment_ID)
--               Measure the number of shipments handled by each carrier.
--   Measure 2 - Average Delivery Days: AVG(Delivery_Days)
--               Measure the average delivery speed of each carrier. Lower
--               average delivery days indicate faster transportation
--               performance.
--   Measure 3 - Average Freight Cost : AVG(Freight_Cost)
--               Compare the average freight cost associated with each carrier.
--   Measure 4 - Total Freight Cost   : SUM(Freight_Cost)
--               Measure the total transportation spend associated with
--               each carrier.
SQL Query - Overall Carrier Performance:
```

```sql
SELECT
    Carrier,
    COUNT(Shipment_ID) AS Total_Shipments,
    ROUND(AVG(Delivery_Days), 2) AS Avg_Delivery_Days,
    ROUND(AVG(Freight_Cost), 2) AS Avg_Freight_Cost,
    ROUND(SUM(Freight_Cost), 2) AS Total_Freight_Cost
FROM Transportation
```

```
GROUP BY Carrier
ORDER BY Avg_Delivery_Days ASC;
```

Expected Output:

```
--   Carrier    | Total Shipments | Avg Delivery Days | Avg Freight Cost | Total Freight Cost
--   UPS        | 1,268           | 4.87              | 10,379.64        | 13,161,379
--   FedEx      | 1,234           | 4.88              | 10,194.05        | 12,579,455
--   BlueDart   | 1,234           | 4.90              | 10,587.59        | 13,065,081
--   DHL        | 1,264           | 4.93              | 10,476.71        | 13,242,565
--   Overall carrier delivery performance was relatively similar across the
--   transportation network. UPS recorded the lowest average delivery time
--   at 4.87 days, while DHL recorded the highest at 4.93 days, a
--   difference of only 0.06 days. FedEx recorded the lowest average
--   freight cost at 10,194.05, while BlueDart recorded the highest at
--   10,587.59. Since network-level averages showed limited
--   differentiation in delivery performance, further analysis was
--   conducted at the transportation-lane level.
```

Analysis 2: Transportation Lane Performance

Business Question:

```
--   Does carrier cost and delivery performance vary across specific
--   Origin-Destination transportation lanes?
```

Business Objective:

```
--   Analyze carrier performance at the transportation-lane level to
--   identify route-specific cost-service trade-offs that may not be
--   visible in overall carrier averages.
```

Desired Output Grain:

```
--   One row represents one Origin + Destination + Carrier combination.
```

SQL Query - Lane-Level Carrier Performance:

```
SELECT
    Origin,
    Destination,
    Carrier,
    COUNT(Shipment_ID) AS Total_Shipments,
    ROUND(AVG(Delivery_Days), 2) AS Avg_Delivery_Days,
    ROUND(AVG(Freight_Cost), 2) AS Avg_Freight_Cost
FROM Transportation
GROUP BY
    Origin,
    Destination,
    Carrier
ORDER BY
    Origin,
    Destination,
    Avg_Delivery_Days ASC;
```

Expected Output:

```
--   Chennai -> Bangalore : BlueDart and UPS recorded the fastest average
--     delivery time (4.87 days); UPS achieved the same average delivery
--     time at a lower average freight cost than BlueDart. FedEx recorded
--     the lowest average freight cost (10,073.53) with a slightly higher
--     average delivery time (5.07 days). Assessment: UPS provides a
--     strong cost-service balance for this lane, while FedEx may suit
--     cost-sensitive shipments.
--   Mumbai -> Delhi : DHL recorded the fastest average delivery time
--     (4.83 days); FedEx followed closely (4.85 days) while maintaining a
--     lower average freight cost (10,270.60) versus DHL (10,665.20).
```

Assessment: FedEx offers a favourable cost-service balance on this

```
--     lane.
```

```
--   Pune -> Hyderabad : FedEx recorded the fastest average delivery time
--     (4.72 days) with an average freight cost of 10,235.99; DHL recorded
--     the lowest average freight cost (10,177.60) but the highest average
--     delivery time (4.95 days). Assessment: FedEx provides a competitive
--     cost-service balance for this transportation lane.
```

Overall Finding: the initial network-level carrier analysis showed

```
--   very similar average delivery performance across carriers; however,
--   analyzing performance at the Origin-Destination-Carrier level revealed
--   route-specific differences in freight cost and delivery speed. Carrier
--   performance should not be evaluated solely using network-wide
--   averages.
```

Business Interpretation:

```
--   A network-wide carrier strategy may overlook route-specific
--   differences in carrier performance. By evaluating carriers at the
--   transportation-lane level, the organization can make more informed
--   carrier allocation decisions based on the required balance between
--   delivery speed, freight cost, shipment urgency, and route-specific
--   performance. This approach can potentially improve transportation
--   cost efficiency while maintaining required service levels.
```

Immediate actions: evaluate carrier performance at the

```
--   transportation-lane level rather than relying solely on network-wide
--   averages, use faster carriers for urgent or service-critical
--   shipments where the cost premium is justified, consider lower-cost
--   carriers for routine shipments where slightly longer delivery times
--   are operationally acceptable, and review carrier allocation
--   strategies for high-volume transportation lanes. Medium-term actions:
--   develop lane-level carrier performance scorecards, monitor average
--   freight cost and delivery time by carrier and route, introduce
--   additional carrier KPIs where data becomes available (e.g. On-Time
--   Delivery %, damage rate, shipment reliability, cost per weight or
--   distance unit), and use Power BI to monitor cost-service trade-offs
--   dynamically.
```

Data Limitations: the analysis compares carriers using the available

```
--   transportation data (Average Delivery Days, Average Freight Cost,
--   Shipment Volume). The dataset does not contain promised delivery
--   dates, actual delivery dates, shipment weight or volume,
--   distance-normalized freight cost, damage or claims data, or On-Time
--   Delivery percentage. Therefore, the analysis should not be
--   interpreted as a complete carrier performance scorecard.
```

END OF FILE: 01_SQL_Analysis.sql